

Tools and Pipelines for HIV-1 Proviral U3 Sequence Analysis

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Introduction

This review outlines bioinformatics tools, pipelines, and methodologies that can be used to achieve Specific Objective 1 of the HIV-1 proviral U3 sequence analysis study: *To extract and annotate U3 sequences from intact near-full-length HIV-1 proviral genomes (FLIP-SEQ and HIV SMRTcap) in the Rakai Cohort, stratified by subtype (A1, D, CRF/URF), and to systematically map binding sites for host transcription factors and G-quadruplexes.*

It covers tools from early standards to current high-throughput applications across the Bash, R, and Python computational ecosystems.

The steps required for this objective are:

- Sequence (FASTQ) preprocessing, quality control
- Genome Assembly
- Biological filtering and intactness assessment
- Multiple sequence alignment
- HIV subtype verification and determination
- Proviral sequence mapping, U3 sequence extraction, and motif identification

1. Raw Data Processing, Quality Control

High-throughput sequencing (HTS) generates large volumes of raw sequence data that contain artefacts, PCR duplicates, and potential host or bacterial contamination. The first step in any bioinformatics pipeline is rigorous quality control (QC) and read processing, to ensure that only high-quality viral reads proceed downstream. Quality assessment evaluates metrics that indicate sequence quality, including quality scores, duplication, GC content, and contamination, and is followed by filtering and trimming against set thresholds.

FastQC

FastQC is used for initial quality assessment of raw sequencing reads, providing the standard first look at data quality before any processing. It runs a set of modular analyses that evaluate basic sequence statistics, per-base and per-sequence Phred quality scores, per-base sequence and GC content, per-base N content, sequence length distribution, GC content distribution, sequence

duplication levels, and overrepresented sequences and adapter content. It accepts standard FASTQ, BAM, or SAM files and generates HTML reports for every metric. It can be run interactively through a desktop application or non-interactively from the command line. It is applied to data from any platform but is most informative for short-read Illumina data, where its per-base quality and adapter checks are most useful. It is a universal diagnostic step in core sequencing facilities worldwide. Its main limitation is the absence of batch processing, since it evaluates one file at a time, which is cumbersome for large cohorts, and it only reports metrics without performing any trimming or modification. FastQC was developed by Simon Andrews at the Babraham Institute (Andrews, 2010), and is written in Java.

MultiQC

MultiQC is used to aggregate quality-control outputs from many samples into a single report, removing the need to review hundreds of individual FastQC results. Its modular architecture sweeps through directory trees to locate and parse log files from FastQC and over 100 other bioinformatics tools, compiling them into one interactive HTML report with dynamic plots. It is also a cross-platform command-line tool and works with data from both short-read and long-read platforms, summarising metrics across an entire cohort regardless of sequencing technology. It is a standard component of modern automated reporting pipelines worldwide, particularly in high-throughput RNA-Seq and viral whole-genome sequencing (WGS) workflows, where it makes cohort-scale quality assessment practical. MultiQC was developed by Phil Ewels at SciLifeLab (Ewels et al., 2016, *Bioinformatics*), and is written in Python.

Trimmomatic

Trimmomatic is used for physical trimming of sequencing reads, removing adapters and low-quality bases before assembly. It applies an ordered series of operations, offering options such as ILLUMINACLIP for adapter removal, SLIDINGWINDOW for quality-based trimming, LEADING and TRAILING base removal, and MINLEN length filtering. It is designed for short-read Illumina data, both single and paired-end FASTQ files and runs from the command line. It has long been a standard preprocessing step in read workflows and offers precise algorithmic control. Its main limitation is that its legacy Java architecture is slower and more memory-intensive than newer tools, which can create bottlenecks on large datasets. Trimmomatic was developed by

Anthony Bolger and colleagues at RWTH Aachen University (Bolger et al., 2014, *Bioinformatics*), and is written in Java.

fastp

fastp is an all-in-one tool that performs quality assessment and preprocessing, including trimming, filtering, and reporting, in a single pass through the data. It handles deduplication, adapter trimming, polyG tail trimming (important for NextSeq and NovaSeq data), and quality filtering simultaneously, and automatically generates JSON and HTML reports. It is a command-line tool with multithreading flags that let it run in parallel across multiple CPU cores on high-performance computing (HPC) clusters. It is well suited to short-read Illumina data, especially from NextSeq and NovaSeq instruments. It provides a fast, scriptable alternative to the FastQC-plus-Trimmomatic workflow for large cohorts. One trade-off is less granular sliding-window trimming: whereas Trimmomatic sets window size and quality threshold independently (for example SLIDINGWINDOW:4:20 trims where a 4-base window drops below Phred 20), fastp bundles sliding-window trimming within broader directional flags such as `--cut_right` and `--cut_mean_quality`, making position-specific trimming less straightforward. fastp was developed by Shifu Chen (Chen et al., 2018, *Bioinformatics*), and is written in C++.

Cutadapt

Cutadapt is used to locate and remove adapter sequences, primers, and other contaminating sequence from high-throughput reads. It uses error-tolerant alignment to find adapters and then clips the matched portions, and also supports quality trimming and length filtering. It runs from the command line and works with short-read Illumina data as well as reads from other platforms. It is widely used as the trimming engine inside larger pipelines such as Trim Galore, and is a reliable, well-documented option where precise adapter removal is needed. Cutadapt was developed by Marcel Martin (Martin, 2011, *EMBnet.journal*), and is written in Python.

Kraken2

Kraken2 is used to detect and filter human host or bacterial contamination, so that non-viral reads are not erroneously assembled alongside the viral genome. It classifies reads taxonomically using exact k-mer matching and memory mapping against a reference database, and outputs text-based classification reports that allow host or contaminant reads to be filtered out. It runs from the

command line and accepts both short- and long-read FASTQ/FASTA data. It is a standard contamination-filtering step viral genome reconstruction. Its main limitation is a substantial RAM footprint, often more than 50 GB, because it loads the full NCBI nucleotide database into working memory for rapid comparison. Kraken2 was developed by Derrick Wood and colleagues at Johns Hopkins University (Wood et al., 2019, *Genome Biology*), and is written in C++.

PacBio SMRT Link

PacBio SMRT Link is the software suite used to process third-generation long-read sequencing data from Pacific Biosciences instruments. Its core function is to transform raw SMRT subreads (.bax.h5 and BAM files) into high-fidelity Circular Consensus Sequence (CCS) reads through repeated sampling and error correction; the ccs tool, with parameters such as --min-passes and --min-rq, handles PacBio's characteristic insertion/deletion error profile. It can be accessed through a browser interface or the command line and is the standard platform for PacBio long-read data. It can span an entire ~9 kb proviral genome in a single read, which is valuable for HIV-SMRTcap datasets. It is proprietary, closed-source, and resource-intensive, requiring substantial compute nodes. SMRT Link was developed by Pacific Biosciences (Pacific Biosciences, 2024), and is written in C++ and Java.

NanoPlot

NanoPlot is used for quality control and visualisation of long-read sequencing data, serving a role for long reads analogous to that of FastQC for short reads. It generates plots and summary statistics for read-length and quality distributions, yield over time, and the relationship between read length and quality, helping assess whether a long-read run met expectations. It runs from the command line as part of the NanoPack toolset. It is designed for PacBio and Oxford Nanopore long-read data and is a common QC step for long-read projects; for this study it complements PacBio SMRT Link on HIV-SMRTcap data. NanoPlot was developed by Wouter De Coster and colleagues (De Coster et al., 2018, *Bioinformatics*), and is written in Python.

2. Genome Assembly

BWA-MEM and Bowtie2

BWA-MEM and Bowtie2 are general-purpose short-read mappers used to align sequencing reads to a reference genome for reference-based assembly, variant calling, and coverage assessment. Both build a memory-efficient index of the reference (a Burrows-Wheeler / FM-index) and then align reads to it; BWA-MEM handles both short and longer reads, while Bowtie2 is optimised for short reads. They both run from the command line and produce SAM or BAM alignments and are designed for short-read Illumina data. They are foundational infrastructure used across virtually all read-mapping workflows, frequently as the mapping step inside HIV assembly pipelines such as SHIVER. BWA-MEM was developed by Heng Li (Li, 2013, *arXiv:1303.3997*; Li and Durbin, 2009, *Bioinformatics*) and Bowtie2 by Ben Langmead and Steven Salzberg (Langmead and Salzberg, 2012, *Nature Methods*); both are written in C/C++.

SAMtools

SAMtools is the standard utility for manipulating sequence alignment files and for indexed retrieval of reference sequences. It sorts, indexes, filters, and converts between the SAM, BAM, and CRAM formats, and its `faidx` command provides instant coordinate-based retrieval of sequences from an indexed FASTA file. It runs from the command line and operates on alignments from any read mapper and on FASTA references, independent of platform. It is essential infrastructure throughout sequencing workflows, including extraction of specific genomic regions such as the U3 segment from aligned data. SAMtools was developed by Heng Li and colleagues (Li et al., 2009, *Bioinformatics*; Danecek et al., 2021, *GigaScience*), and is written in C.

SHIVER

SHIVER (Sequences from HIV Easily Reconstructed) is used to assemble highly diverse viral genomes, particularly HIV-1, from short sequencing reads. Rather than mapping reads to a single fixed generic reference, which can introduce bias given HIV's high genetic diversity, SHIVER dynamically builds a sample-specific custom reference tailored to each sample and maps reads against it, improving reconstruction accuracy. It reconstructs near-full-length genomes in FASTA format and also outputs base frequencies, consensus sequences, and minority-variant information, with all parameters controlled through a configuration file. It is a suite of Bash and Python scripts run from the command line using two key commands, `shiver_init.sh` to prepare the reference and

shiver_align.sh to map paired-end FASTQ reads. It is optimised for Illumina short-read data; although designed for HIV-1 and HCV, it has been applied successfully to other viruses such as respiratory syncytial virus. Its main limitation is a steep learning curve arising from its Docker-based setup and stringent input requirements. SHIVER was developed by Chris Wymant and Christophe Fraser at Oxford University (Wymant et al., 2018, *Virus Evolution*), and is written in Bash and Python.

SPAdes

SPAdes is a de novo assembler used to reconstruct genomes from short reads without a reference, which is useful for capturing divergent or recombinant regions that reference-based methods may miss. It constructs and resolves a de Bruijn graph across multiple k-mer sizes to produce contigs, and provides specialised modes such as metaSPAdes for mixed populations and coronaSPAdes for coronavirus data. It runs from the command line and works best with Illumina short reads, and also offers hybrid modes that incorporate long reads to improve scaffolding. It is one of the most widely used assemblers in microbial and viral genomics and serves as the assembly engine in numerous automated pipelines and web platforms, including Genome Detective. SPAdes was developed by Anton Bankevich and colleagues at the Center for Algorithmic Biotechnology, St. Petersburg (Bankevich et al., 2012, *Journal of Computational Biology*), and is written in C++ and Python.

Genome Detective

Genome Detective is a web-based cloud pipeline used for rapid viral NGS assembly without command-line or programming expertise. It runs established tools such as DIAMOND and SPAdes on its backend; users upload FASTQ files through a graphical interface and receive annotated FASTA assemblies and HTML reports. It is most suited to short-read Illumina data, with some long-read capability. Its limitations are cloud-processing bottlenecks during high traffic and a lack of the fine-grained algorithmic control needed for detailed proviral intactness metrics. Genome Detective was developed by the group of Tulio de Oliveira (Cleemput et al., 2020, *Bioinformatics*), and is provided as a hosted web service.

Recommendation for this study

For Illumina (FLIP-Seq) reads, run fastp for trimming and quality filtering (it scales well across a cohort and handles polyG artefacts from NovaSeq), then Kraken2 for host and contaminant removal, then SHIVER for assembly. SHIVER's sample-specific reference avoids the reference bias that fixed-reference mappers introduce on divergent subtype A1 and D genomes and recombinants.

For PacBio (HIV-SMRTcap) reads, generate CCS/HiFi reads with the SMRT Link ccs tool and assess run quality with NanoPlot. Aggregate per-sample metrics across both platforms with MultiQC.

3. Biological Filtering and Intactness Assessment

HIV-1 integration is highly error-prone, and a critical biological filtering step is required because over 90% of integrated proviruses are genetically defective owing to APOBEC3G/F-induced hypermutation, large internal deletions, premature stop codons, or structural defects. The proviral landscape of the Rakai cohort across subtypes A1, D, and recombinants has been characterised in detail (Lee et al., 2024, *Nature Communications*). Rigorous intactness QC is essential so that downstream transcription factor binding site (TFBS) analysis is performed only on functional proviruses capable of latency reversal.

3.1 Hypermutation and Defect Inference

Poplars (Hypermur 3)

Poplars (Hypermur 3) is used to detect APOBEC3G/F-induced G-to-A hypermutation, the earliest recognised biological defect in HIV sequences, which occurs in specific dinucleotide contexts (for example GpA to ApA). It compares aligned FASTA sequences to a reference and applies Fisher exact tests to detect hypermutation statistically, based on user-defined sequence contexts, modernising the classic LANL Hypermur web algorithm for high-throughput use. It is run from the command line (for example `python -m poplars hypermur`), which bypasses LANL web-server upload limits and allows automated filtering on HPC. It requires sequences to be pre-aligned to a reference, operates on aligned FASTA sequences from any platform, and is a standard hypermutation screen in proviral pipelines. Poplars was developed by Art Poon at Western University (Poon et al., 2025, *GitHub*), and is written in Python.

3.2 Structural Intactness Classification

HIVSeqinR

HIVSeqinR is a comprehensive pipeline used to classify near-full-length proviruses as intact or defective, and is the established standard for this task in East African contexts. It processes FASTA assemblies to check for large internal deletions, premature stop codons, and structural defects in the Major Splice Donor (MSD) site or Rev Response Element (RRE), and outputs detailed CSV summaries and partitioned FASTA files of intact versus defective genomes. It runs from the command line and executes natively in R on FASTA assemblies for either short- or long-read sequences. Its principal strength is explicit validation on the Rakai cohort's subtypes A1 and D, which makes it a strong surrogate for the Quantitative Viral Outgrowth Assay (QVOA). HIVSeqinR was developed collaboratively by Johns Hopkins University and the Rakai Health Sciences Program (Lee et al., 2021, *Viruses*), and is written in R. Running in R makes it slower than Python alternatives.

HIVIntact

HIVIntact is used for automated, high-throughput inference of proviral intactness in large cohort studies where scalability is the priority. It relies on Open Reading Frame (ORF) parsing to identify structural defects and outputs CSV matrices at a fraction of the computational time of R-based tools. It runs from the command line (for example `hivintact --input seqs.fasta`) and operates on FASTA files; its underlying heuristics were optimised mainly for subtype B genomes and is suited for very large cohorts. Its main limitation is that subtype-B optimisation can bias defect classification when applied to diverse African cohorts, so results on subtype A1 and D data should be interpreted with caution. HIVIntact was developed by I. A. Wright and colleagues at the Blankson Lab (Wright et al., 2021, *Retrovirology*), and is written in Python.

ProSeq-IT

ProSeq-IT (Proviral Sequence Annotation and Intactness Test) is a web tool used to annotate proviral genomic features, infer sequence intactness, and check 5'/3' LTR symmetry. It aligns sequences to subtype references, annotates genomic features, and flags defects such as deletions, hypermutation, frameshifts, and inversions. It is accessed through the web interface of the National Cancer Institute's HIV-1 Proviral Sequence Database (PSD): users select a subtype, upload FASTA sequences, and submit.

It currently supports only subtypes B and C and requires FASTA input without ambiguous nucleotides. It is used for proviral annotation and intactness checks within the PSD ecosystem. The restriction to subtypes B and C limits its applicability to this cohort. ProSeq-IT is provided by the NCI HIV-1 Proviral Sequence Database (Shao et al., 2020, *AIDS Research and Human Retroviruses*), and is web-based.

Recommendation for this study

Two-stage filter; first screen for APOBEC-driven hypermutation with Poplars (Hypermut 3), then classify structural intactness with HIVSeqinR.

4. Multiple Sequence Alignment (MSA)

Following extraction of intact proviruses, sequences must be placed on a common coordinate system. HIV-1 shows profound inter-clade nucleotide divergence (up to 25%), so a highly accurate multiple sequence alignment (MSA) anchored to a global reference such as HXB2 is required to establish consistent genomic coordinates for precise downstream mapping of the U3 regulatory region.

MAFFT (L-INS-i)

MAFFT is a global standard widely used, high-accuracy multiple sequence aligner; its iterative-refinement L-INS-i mode is intended for the most accurate alignment of highly divergent sequences. It uses Fast Fourier Transform (FFT) to identify homologous regions rapidly; the L-INS-i algorithm (for example `mafft --localpair --maxiterate 1000 in.fasta > out.fasta`) repeatedly recalculates distance matrices and applies local pairwise refinement to maximise accuracy. It runs from the command line and works on assembled genomes from any platform. Its main limitation is computational cost: L-INS-i scales poorly and can require hours on an HPC node for thousands of full-length genomes. MAFFT was developed by Kazutaka Katoh at Osaka University (Katoh and Standley, 2013, *Molecular Biology and Evolution*), and is written in C.

MUSCLE

MUSCLE is a general-purpose aligner used as a faster, practical alternative to MAFFT for nucleotide and protein sequences. It uses iterative progressive alignment with guide-tree refinement until convergence; the latest version (Muscle5) produces an ensemble of alternative alignments to assess confidence. It runs from the command line and handles nucleotide and protein

sequences from any platform. It is a long-standing phylogenetics default and is embedded as the alignment engine in packages such as MEGA. Its accuracy on highly divergent sequences typically trails MAFFT L-INS-i, which remains preferred in those cases. MUSCLE was developed by Robert C. Edgar (Edgar, 2004, *Nucleic Acids Research*; Edgar, 2022, *Nature Communications*), and is written in C++.

Clustal Omega

Clustal Omega is used to align very large sequence sets efficiently available as a command-line tool and through web servers. It builds a fast guide tree using modified mBed clustering and then performs progressive alignment with hidden Markov model profiles. It scales to large numbers of sequences from any platform; for a small number of highly divergent sequences, its accuracy can trail iterative refiners such as MAFFT L-INS-i. Clustal Omega was developed by Fabian Sievers and colleagues at University College Dublin (Sievers et al., 2011, *Molecular Systems Biology*), and is written in C/C++.

MEGA

MEGA (Molecular Evolutionary Genetics Analysis) is an integrated package used for sequence alignment, phylogenetic tree building, and evolutionary analysis in a single environment. It bundles alignment (via MUSCLE and ClustalW), substitution-model selection, distance estimation, and tree inference behind a unified interface. It provides both a graphical desktop application and a command-line version (MEGA-CC). It handles nucleotide and protein sequences from any platform and is best suited to interactive, exploratory analysis rather than very large cohort-scale datasets. It is popular among researchers who value accessibility and interactivity, and is widely used for teaching and for quick exploratory alignment and tree building. MEGA was developed by Sudhir Kumar, Koichiro Tamura, and colleagues (Tamura, Stecher and Kumar, 2021, *Molecular Biology and Evolution*), and is written in C++ with a graphical front end.

Biopython (AlignIO) and Bioconductor (msa)

Biopython (AlignIO) in Python and Bioconductor (msa) in R are scripting libraries used to run alignment tools and parse alignments natively within analysis scripts. They act as programmatic wrappers around command-line aligners and load sequence sets into the language environment as objects and are used inside Python and R scripts respectively. They are platform-agnostic but memory-heavy; loading entire sequence dictionaries frequently causes memory exhaustion, so

they are best suited to small, localised alignments, with direct command-line execution preferred for cohort-scale data. They are common where alignment must be embedded in a larger scripted workflow. Biopython was developed by the Biopython community (Cock et al., 2009, *Bioinformatics*) and is written in Python; the msa package was developed by Bodenhofer and colleagues (Bodenhofer et al., 2015, *Bioinformatics*) and is written in R and C++.

Minimap2

Minimap2 is the standard command-line tool for rapid initial mapping of long reads. It is splice-aware and uses minimizer-based seeding optimised for noisy long reads, with presets such as -ax asm20 for highly divergent sequences or -x map-pb, producing SAM, BAM, or PAF output. It runs from the command line and accepts FASTA or FASTQ input and is the industry standard for long-read mapping. It is optimised for PacBio and Oxford Nanopore long reads, such as HIV-SMRTcap contigs. As a fast general-purpose mapper, it can lack the fine base-pair resolution needed for the highly variable regulatory motifs of the short U3 region, which is why MAFFT (L-INS-i) is preferred for the final alignment. Minimap2 was developed by Heng Li (Li, 2018, *Bioinformatics*), and is written in C.

HIVAlign and Gene Cutter (LANL)

HIVAlign and Gene Cutter, from the Los Alamos National Laboratory (LANL) HIV Sequence Database, are HIV-specific web tools used to align sequences to curated HIV references in a biologically aware way. They use HIV-aware, codon- and structure-informed methods that keep biologically meaningful positions aligned across divergent clades. They are accessed through a web interface, accept FASTA input from any platform and are useful for HIV-specific alignment and for cutting sequences into defined genome regions. They are provided by the LANL HIV Sequence Database (Kuiken et al., 2003, *AIDS Reviews*).

Recommendation for this study

MAFFT in L-INS-i mode anchored to HXB2 given its iterative refinement which gives the highest base-pair-level accuracy across subtype A1, D, and recombinant sequences, important for accurate downstream coordinate mapping of the short U3 motifs. Minimap2 is good for rapid initial placement against the reference while MEGA is useful for interactive exploration and quick tree building without scripting, but should not replace MAFFT for the final cohort-scale alignment on which coordinate mapping depends.

5. Subtype and Recombination Detection

HIV-1 transcriptional regulation is strongly clade-specific. Stratifying the aligned sequences into subtypes (for example A1 and D) and recombinant forms (CRF/URF) is essential so that downstream TFBS abundance analyses are not confounded by phylogenetic background.

5.1 Recombination and Pure Subtyping

jpHMM

jpHMM (jumping profile Hidden Markov Model) is used to resolve recombination by assigning each genomic segment to a parental subtype and calculating precise breakpoints. It compares query FASTA sequences against profile alignments of the major subtypes, allowing the model to jump between profiles, and outputs breakpoint locations with detailed text summaries and PDF subtype-mosaic maps. It is accessed through a web server, with command-line execution wrapped in Perl/Python, and accepts FASTA input. It works across all platforms and is the reference standard for accurate breakpoint detection. Its main limitation is that web-server execution becomes a bottleneck for very large datasets. jpHMM was developed by Anne-Kathrin Schultz and colleagues at the University of Goettingen (Schultz et al., 2006, *Nucleic Acids Research*), and is written in C++ with Perl and Python wrappers.

COMET

COMET is used for rapid subtype classification of pure clades and is accessed through a web interface or an API and returns XML or CSV classifications. It uses context-based statistical models (variable-order Markov chains) rather than phylogenetic tree construction, classifying subtypes almost instantaneously and accepts FASTA data from any platform. It is a fast first-pass classifier however, its statistical approximations are unreliable for resolving the exact breakpoints of the complex recombinant URFs common in African cohorts, so it works best paired with a breakpoint-detection method. COMET was developed by Daniel Struck and colleagues (Struck et al., 2014, *Nucleic Acids Research*), and is maintained as a web/API service.

REGA HIV-1 Subtyping Tool v3

The REGA HIV-1 Subtyping Tool version 3 is a phylogenetics-based classifier used to assign both pure subtypes and recombinant forms for FASTA input from any platform.. It builds a phylogenetic tree against references and uses bootscanning to detect recombination, identifying subtypes A to D, F to H, J, and K, plus CRFs and URFs, with supporting evidence. It is accessed

through a web interface on the Genome Detective platform, with batch output in CSV, XML, or Excel. In head-to-head comparisons it performed well for pure subtypes and, alongside COMET, identified most CRFs, making it a useful complement to jpHMM for datasets rich in East African recombinants; its phylogenetic basis gives more interpretable results than purely statistical methods. REGA v3 was developed at the Rega Institute, KU Leuven, by Pineda-Pena and colleagues (Pineda-Pena et al., 2013, *Infection, Genetics and Evolution*), and operates as a web service.

RIP

RIP (Recombinant Identification Program) is used as a quick visual screen for inter-subtype recombination. It slides a window across the query sequence and calculates similarity to a panel of subtype references at each position, highlighting which subtype the sequence most closely matches so that breakpoints become apparent. It is accessed through the LANL HIV Database web interface and accepts FASTA input. It works across platforms and is a convenient first-pass visual screen for recombination, used alongside more rigorous breakpoint callers. RIP was developed at the LANL HIV Sequence Database (Siepel et al., 1995, *AIDS Research and Human Retroviruses*).

SCUEAL

SCUEAL (Subtype Classification Using Evolutionary ALgorithms) is used for fine-grained recombination analysis and subtype assignment. It uses a genetic algorithm within HyPhy to search possible breakpoint configurations and place the query within a reference phylogeny. It runs within HyPhy and is available through the Datamonkey web platform. It accepts FASTA input from any platform. It produced competitive, fine-grained recombination calls in evaluations, at a higher computational cost than fast statistical methods. SCUEAL was developed by Sergei L. Kosakovsky Pond and colleagues (Kosakovsky Pond et al., 2009, *PLoS Computational Biology*).

Stanford HIVdb

The Stanford HIV Drug Resistance Database (HIVdb / Sierra) is primarily a drug-resistance interpretation system that also reports a subtype assignment as part of its routine analysis. It is accessed through a web interface or a programmatic service and is convenient for secondary subtype confirmation; it was not designed for comprehensive recombination analysis. Stanford HIVdb was developed by Robert Shafer and colleagues at Stanford University (Tang et al., 2012, *Intervirology*).

5.2 Phylogenetic Inference and Molecular Epidemiology

IQ-TREE 2

IQ-TREE 2 is used to confirm subtype assignments with robust statistical support through maximum-likelihood (ML) phylogenetics. It infers ML trees from alignments, with -m MFP for automatic ModelFinder substitution-model selection and -B 1000 for ultrafast bootstrapping, producing NEWICK or NEXUS trees. It runs from the command line and accepts FASTA or PHYLIP alignments. It works on aligned sequences from any platform and is a modern standard for phylogenetic inference. It demands significant HPC wall-time for large alignments. IQ-TREE 2 was developed by Bui Quang Minh and colleagues (Minh et al., 2020, *Molecular Biology and Evolution*), and is written in C++.

RAxML and FastTree

RAxML and FastTree are alternative maximum-likelihood tree builders used alongside or in place of IQ-TREE 2. FastTree uses approximate ML to build trees very quickly, while RAxML performs more rigorous ML estimation with thorough optimisation. Both run from the command line on aligned sequences from any platform. FastTree is well suited to rapid initial exploration of large alignments; RAxML is suited to more rigorous final trees. Together they provide a useful speed/accuracy trade-off complementing IQ-TREE 2. RAxML was developed by Alexandros Stamatakis (Stamatakis, 2014, *Bioinformatics*) and FastTree by Morgan Price and colleagues (Price et al., 2010, *PLoS ONE*); both are written in C/C++.

ggtree

ggtree is used to visualise and annotate phylogenetic trees for publication, built on the ggplot2 grammar of graphics, it parses NEWICK and IQ-TREE output and maps clinical or molecular covariates (such as viral load or intactness) directly onto branches and tips. It is an R/Bioconductor package used within R scripts and works with tree files from any inference tool. It is a standard for publication-ready annotated trees and integrates seamlessly into R workflows, although the underlying tree must be generated by a separate tool first. ggtree was developed by Guangchuang Yu (Yu et al., 2017, *Methods in Ecology and Evolution*), and is written in R.

FigTree and iTOL

FigTree and iTOL are interactive graphical tools used to display and lightly annotate phylogenetic trees without scripting. FigTree is a desktop application; iTOL is a web-based platform with richer

annotation options. Both accept standard tree formats from any source and are convenient for quick figures and presentation graphics. FigTree was developed by Andrew Rambaut at the University of Edinburgh (Rambaut, 2018) and iTOL by Ivica Letunic and Peer Bork (Letunic and Bork, 2021, *Nucleic Acids Research*).

Recommendation for this study

jpHMM as the core recombination caller: it resolves the A1/D and CRF/URF breakpoints that COMET cannot. Confirm subtype calls with a maximum-likelihood phylogeny in IQ-TREE 2 (-m MFP -B 1000). Use COMET and REGA v3 for rapid first-pass classification of pure subtypes and a RIP scan as a quick visual recombination screen, and render the final annotated trees with ggtree.

6. Proviral Mapping, U3 Region Extraction, and Motif Identification

With sequences aligned and subtyped, the regulatory domain of the LTR, the U3 region (spanning approximately 454 base pairs relative to the transcription start site), must be precisely mapped and extracted. Then we locate and quantify binding sites for key host transcription factors (for example NF- κ B, SP1, NFAT) and structural motifs such as G-quadruplexes.

6.1 Coordinate Mapping and Extraction

Biopython (SeqIO) and Bioconductor (Biostrings)

Biopython (SeqIO) in Python and Bioconductor (Biostrings) in R are used to extract subsequences, such as the U3 region, based on alignment coordinates within custom scripts used in Python and R scripts respectively. They load sequences as string objects and slice them by coordinate, giving precise, programmable control over extraction. They are platform-agnostic but slow and memory-intensive on massive batch FASTA files. They are common where extraction must be embedded in a larger scripted workflow; performance degrades on large batches. Biopython was developed by the Biopython community (Cock et al., 2009, *Bioinformatics*) and is written in Python; Biostrings was developed by Pages and colleagues (Pages et al., 2024, Bioconductor) and is written in R and C.

SeqKit

SeqKit is used for fast, memory-efficient, coordinate-based extraction and manipulation of FASTA/FASTQ files. It uses optimised commands (for example `seqkit subseq -r start:end in.fasta`) and utilities such as `fx2tab`, streaming through large files without loading them entirely into RAM.

It is a cross-platform command-line toolkit, handles massive batch alignment FASTA files and is a standard utility across modern bioinformatics groups for instantaneous coordinate-based extraction. SeqKit was developed by Wei Shen (Shen et al., 2016, *PLOS ONE*), and is written in Go.

bedtools and samtools faidx

bedtools and samtools faidx are used to retrieve and manipulate sequence by coordinate, which is convenient when U3 coordinates and motif positions are managed as interval files. samtools faidx indexes a FASTA file and retrieves sequences by coordinate instantly; bedtools getfasta extracts sequences from intervals defined in a BED file, and bedtools can also intersect or merge genomic features. Both run from the command line, work across platforms on indexed FASTA and interval files and are widely used wherever genomic regions are handled as coordinate intervals. bedtools was developed by Aaron Quinlan and Ira Hall (Quinlan and Hall, 2010, *Bioinformatics*) and is written in C++; samtools faidx is part of the SAMtools suite (Li et al., 2009, *Bioinformatics*) and is written in C.

LANL Gene Cutter

The LANL Gene Cutter is an HIV-specific web tool used to cut aligned HIV sequences into defined genome regions (including the LTR/U3) in a codon-aware manner. It aligns sequences to HIV references and extracts named regions, is accessed through the LANL HIV Sequence Database web interface, and accepts FASTA input from any platform. It is provided by the LANL HIV Sequence Database (Kuiken et al., 2003, *AIDS Reviews*).

6.2 Motif and Non-B DNA Identification

MEME Suite (FIMO and MCAST)

The MEME Suite tools FIMO and MCAST are used to map transcription factor binding sites: FIMO scans for individual motif occurrences, and MCAST detects clusters of nearby motifs (such as the NF- κ B/SP1 enhanceosome). FIMO scans FASTA sequences using Position Weight Matrices (PWMs) against Markov background models, calculating log-odds scores and p-values (for example `fimo --thresh 1e-4 jasper_motifs.meme U3_regions.fasta`); MCAST models the spatial proximity of TF clusters. Both run from the command line or through a web interface and take FASTA and motif-matrix input. They work on sequences from any platform and are a long-standing reference standard for motif scanning. The suite outputs TSV and GFF3 files that require

parsing before statistical analysis in R. The MEME Suite was developed by Timothy Bailey and colleagues (Grant et al., 2011, *Bioinformatics*; Bailey and Noble, 2003, *Bioinformatics*), and is written in C.

MOODS

MOODS is used for very fast PWM-based motif scanning within automated Python pipelines. It uses Aho-Corasick automaton pattern matching to scan FASTA sequences against JASPAR matrices, returning coordinates as native Python objects. It is written in C++ with a native Python API and is used inside Python scripts. It operates on sequences from any platform and excels where high-throughput speed is critical. Its limitations are that it lacks the spatial cluster modelling of MCAST and does not produce polished reports. MOODS was developed by Pasi Rastas and colleagues (Korhonen et al., 2009, *Bioinformatics*), and is written in C++ with a Python API.

HOMER

HOMER (Hypergeometric Optimization of Motif EnRichment) is used for motif discovery and for scanning sequences for known motifs. It performs de novo and known-motif enrichment analysis using a hypergeometric / cumulative-binomial framework and a curated motif library, and can also annotate motif positions. It is a command-line suite of programs and scripts and works on sequence sets from any platform, widely used in regulatory-genomics and ChIP-seq analysis and provides an alternative motif-scanning route to the MEME Suite. HOMER was developed by Christopher Benner and colleagues (Heinz et al., 2010, *Molecular Cell*).

TFBSTools

TFBSTools is used to scan for transcription factor binding sites entirely within R, producing results that integrate directly into downstream statistical analysis. Using S4 classes, it scans DNASTringSet objects against curated JASPAR matrices and stores motif predictions as GRanges objects, avoiding intermediate text files; it is typically paired with the JASPAR2024 database. It is an R/Bioconductor package used within R scripts and works on sequences from any platform; performance is bounded by R's memory rather than compiled speed. It is well suited to workflows that feed motif predictions straight into Generalized Linear Mixed Models (GLMMs), bridging sequence analysis and biostatistics. TFBSTools was developed by Ge Tan and Boris Lenhard (Tan and Lenhard, 2016, *Bioinformatics*), and is written in R.

gquad and Non-BDNAP

The gquad R package (and its associated web tool, Non-BDNAP) is used to predict intramolecular G-quadruplexes and other non-B DNA motifs. Using regular-expression rules based on G-tract and loop-length configurations, it processes DNA strings to predict G-quadruplexes and reports coordinates and formation probabilities as data frames. Because the Non-BDNAP web tool restricts uploads to 200 KB, the native gquad R package is preferable for cohort-scale data. Overlaying its G4 coordinates with TFBS maps from TFBSTools gives a more complete functional picture of the latent HIV-1 promoter. gquad was developed by Olusola (Olusola, 2020, R package), and is written in R.

QGRS Mapper and pqsfinder

QGRS Mapper and pqsfinder are used to predict potential G-quadruplex-forming sequences and to confirm G4 predictions. QGRS Mapper scores G-quadruplex-forming sequences based on tetrad and loop configurations; pqsfinder uses an exhaustive, imperfection-tolerant search that tolerates bulges and mismatches. QGRS Mapper is a web server while pqsfinder is an R/Bioconductor package used within R. They are commonly used to cross-check G-quadruplex predictions, and pqsfinder works well alongside gquad as a consistency check in the U3 region. QGRS Mapper was developed by Oguzhan Kikin and colleagues (Kikin et al., 2006, *Nucleic Acids Research*); pqsfinder was developed by Jiri Hon and colleagues (Hon et al., 2017, *Bioinformatics*), and is written in R/C++.

Motif databases

The PWMs scanned by these tools are drawn from curated collections. **JASPAR 2024** is the open-access standard and is recommended here, version-locked for reproducibility; **HOCOMOCO** and **TRANSFAC** are viable alternatives (Castro-Mondragon et al., 2024, *Nucleic Acids Research*).

Recommendation for this study

Extract the U3 region with SeqKit (seqkit subseq) from the HXB2-anchored alignment; samtools faidx and bedtools getfasta are equivalent if coordinates are held as interval files. Map binding sites with FIMO using JASPAR 2024 PWMs which provides p-value-backed log-odds scoring. Use TFBSTools in parallel, because its GRanges output feeds directly into the downstream GLMM/mixed-effects models in R without text parsing. Predict G-quadruplexes with gquad and

confirm a subset with pqsfinder, then overlay the G4 and TFBS coordinates to build an integrated regulatory map of the U3 promoter.

End-to-End Pipeline for this objective

The following pipeline summarises the recommended tools for each step of Objective 1.

1. **Quality control and assembly.** Illumina (FLIP-Seq): fastp → Kraken2 → SHIVER. PacBio (HIV-SMRTcap): SMRT Link (ccs) → NanoPlot. Aggregate QC across both platforms with MultiQC.
2. **Biological filtering and intactness.** Poplars (Hypermut 3) for hypermutation, then HIVSeqinR for structural intactness (validated on Rakai A1/D). Restrict downstream analysis to HIVSeqinR-confirmed intact proviruses; optionally cross-check a subset with HIVIntact.
3. **Multiple sequence alignment.** MAFFT L-INS-i anchored to HXB2 for the precision alignment; Minimap2 for initial placement of PacBio contigs.
4. **Subtype and recombination detection.** jpHMM as the core recombination caller with IQ-TREE 2 (-m MFP -B 1000) confirmation; COMET and REGA v3 for first-pass pure-subtype calls; ggtree for figures.
5. **U3 extraction and motif mapping.** SeqKit to extract U3; FIMO with JASPAR 2024 for TFBS mapping (and TFBSTools for R-native integration into the statistical models); pqsfinder and gquad for G-quadruplexes; overlay the G4 and TFBS maps.

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